ACYCLICITY OF THE KERNEL PERSISTS UNDER ARBITRARY BASE CHANGE IN THE T DIRECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0FZH: sources/stacks-chow/tags.txt, source lines 19952–19952. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SB: sources/stacks-chow/tags.txt, source lines 2332–2332. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0FZE: sources/stacks-chow/tags.txt, source lines 19949–19949. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let R, X, Y, and $\mathcal{K}$ be as in Lemma 0FZH part (2). Then for any scheme T over R we have*

$$R^q pr_{13,*}(pr_{12}^*\mathcal{F} \otimes_{\mathcal{O}_{T\times_R X\times_R Y}} pr_{23}^*\mathcal{K}) = 0$$

for $\mathcal{F}$ quasi-coherent on $T \times_R X$ and $q > 0$.

Proof. The question is local on T hence we may assume T is affine. In this case we can consider the diagram

$$\begin{array}{ccccc} T \times_R X & \longleftarrow & T \times_R X \times_R Y & \longrightarrow & T \times_R Y \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & X \times_R Y & \longrightarrow & Y \end{array}$$

whose vertical arrows are affine. In particular the pushforward along $T \times_R Y \to Y$ is faithful and exact (Cohomology of Schemes, Lemma 01XC and Morphisms, Lemma

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01SB). Chasing around in the diagram using that higher direct images along affine morphisms vanish (see reference above) we see that it suffices to prove

$$R^q\mathrm{pr}_{2,*}(\mathrm{pr}_{23,*}(\mathrm{pr}_{12}^*\mathcal{F} \otimes_{\mathcal{O}_{T\times_R X\times_R Y}} \mathrm{pr}_{23}^*\mathcal{K})) = R^q\mathrm{pr}_{2,*}(\mathrm{pr}_{23,*}(\mathrm{pr}_{12}^*\mathcal{F}) \otimes_{\mathcal{O}_{X\times_R Y}} \mathcal{K}))$$

is zero which is true by assumption on $\mathcal{K}$. The equality holds by Remark 0FZE. $\square$